

PROJECT TITLE

BANK ACCOUNT MANAGEMENT SYSTEM

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INTRODUCTION

- A simple Java banking system that demonstrates a simple banking system using Java and OOP concepts.
- It provides a clear structure for modelling real-world bank account operations.

PROBLEM STATEMENT

- To design a banking system capable of handling deposits, withdrawals, overdraft limits, and transaction history.
- Basic banking tasks require automation to ensure accuracy and easy tracking of account transactions.

OBJECTIVES OF THE SYSTEM

- Automate basic banking operations.
- Maintain clear and accurate transaction records.
- Apply rules like minimum balance and overdraft limits.
- Handle different account types effectively.

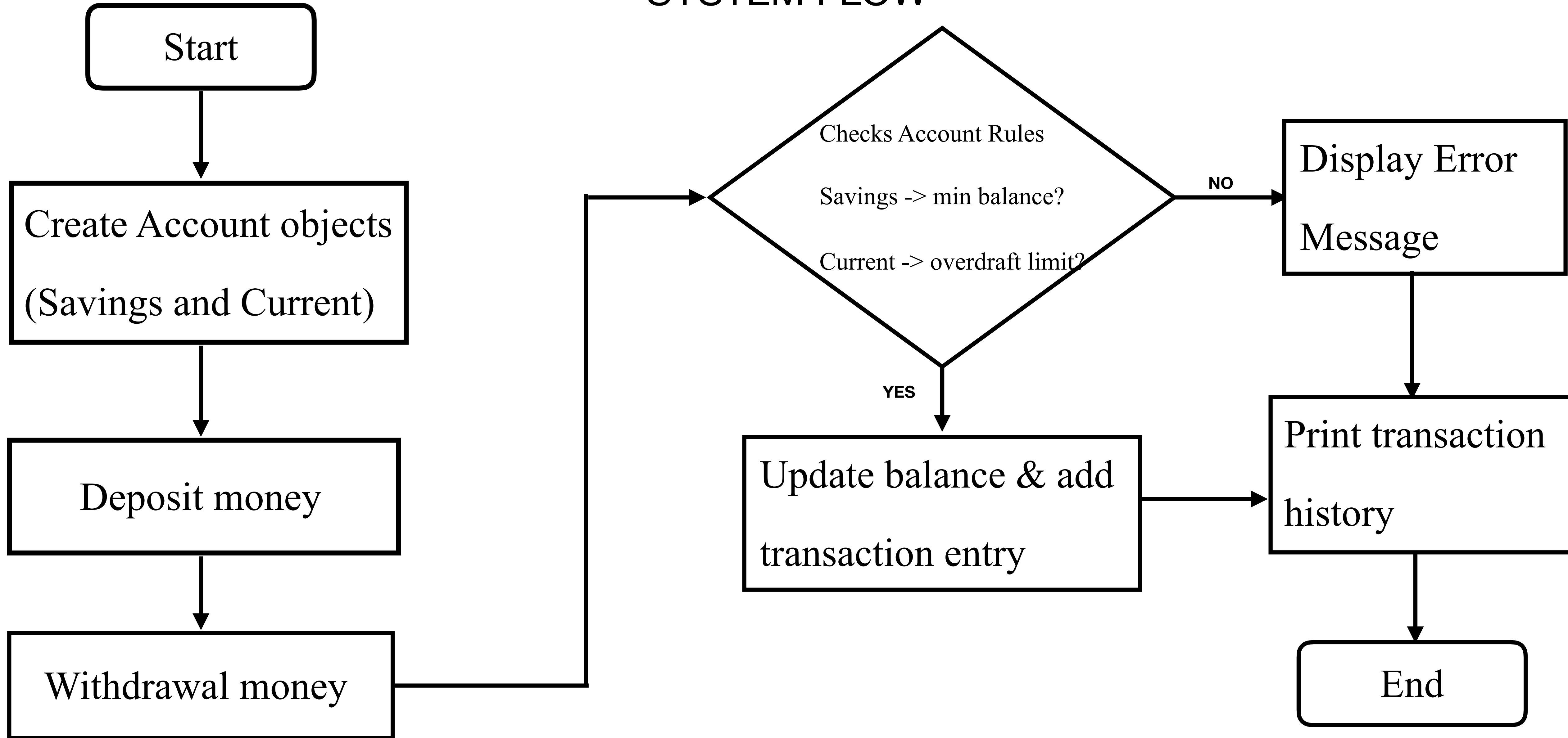
TECHNOLOGIES USED

- **Java Programming Language** - Core language used to build the entire system.
- **Object-Oriented Programming (OOP)** - Concepts like inheritance, abstraction, encapsulation, and polymorphism.
- **Java Collections Framework** - ArrayList is used to store transaction history.

OOPS CONCEPT USED

- Inheritance
- Abstraction
- Polymorphism
- Encapsulation

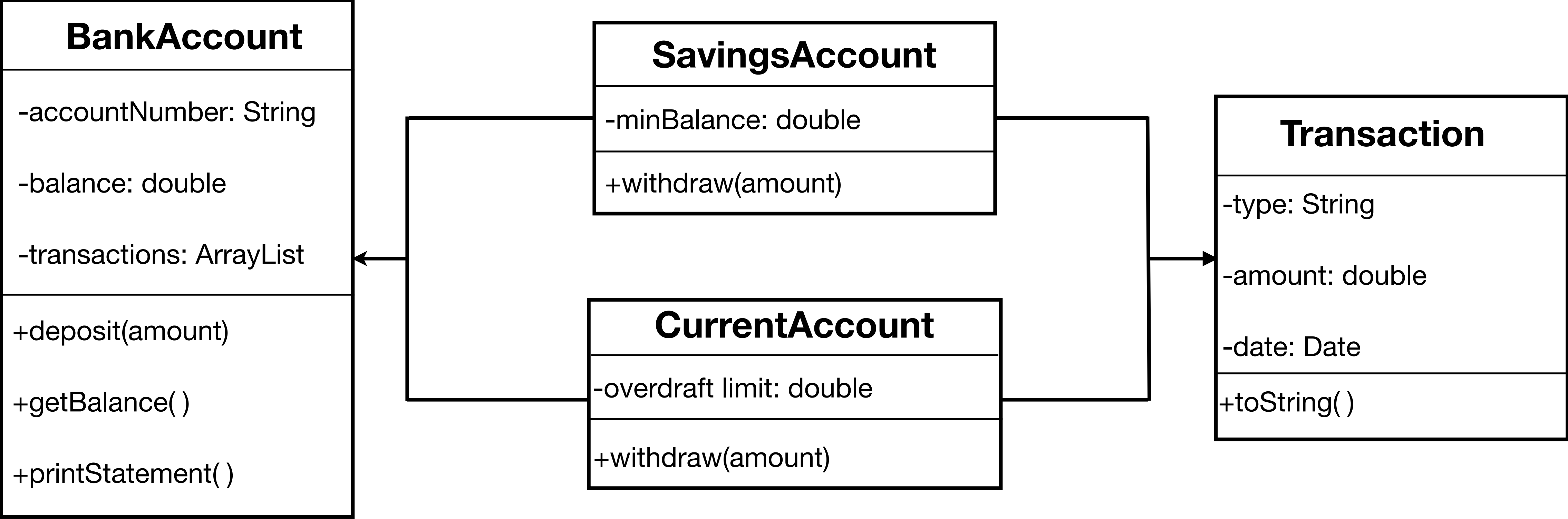
SYSTEM FLOW



FEATURES IMPLEMENTED

- Deposit and withdrawal operations
- Balance rule checking
- Transaction recording
- Statement generation

CLASS DIAGRAM



PROGRAM OUTPUT

```
tharikhas@Mac ~ % /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-25.jdk/Contents/Home/bin/java
--enable-preview -XX:+ShowCodeDetailsInExceptionMessages -cp /private/var/folders/cj/l8zrxc8s6tg24
byxnh1kcdth0000gn/T/vscodesws_73062/jdt_ws/jdt.ls-java-project/bin BankSystem
---- Create Savings Account ----
Enter Savings Account Number: SA1001
Enter Initial Balance: 10000

---- Create Current Account ----
Enter Current Account Number: CA2001
Enter Initial Balance: 15000

---- Savings Account Operations ----
Enter deposit amount: 5000
Deposited : 5000.0 | New Balance: 15000.0
Enter withdrawal amount: 3000
Saving withdrawal successful.New balance: 12000.0

----- Transaction statement for account: SA1001-----
Thu Nov 20 16:10:19 IST 2025 | Deposit| Amount: 5000.0
Thu Nov 20 16:10:27 IST 2025 | Withdrawal| Amount: 3000.0
Current Balance: 12000.0
-----

---- Current Account Operations ----
Enter deposit amount: 4000
Deposited : 4000.0 | New Balance: 19000.0
Enter withdrawal amount: 2000
Current Account Withdrawal Successful. New Balance: 17000.0

----- Transaction statement for account: CA2001-----
Thu Nov 20 16:10:39 IST 2025 | Deposit| Amount: 4000.0
Thu Nov 20 16:10:45 IST 2025 | Withdrawal| Amount: 2000.0
Current Balance: 17000.0
-----
```

CONCLUSION

- The system effectively handles basic banking operations using OOP principles and ensures secure, rule-based transactions.

FUTURE ENHANCEMENTS

- Add login system
- Add interest calculation
- Use database storage
- Provide GUI interface

THANK YOU!